

September 2024 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points**

- (a) An experiment consists of drawing 3 cards in succession from a well-shuffled ordinary deck of cards. Let A_i denote the event of “*obtaining king on i^{th} draw*”, for $i = 1, 2, 3$. Further, let A_i^c denote the complement of event A_i , for $i = 1, 2, 3$. State clearly in words the meaning of each of the following expressions:

- i. $\Pr(A_1 \cap A_2^c)$,
- ii. $\Pr(A_1 \cup A_2)$,
- iii. $\Pr(A_1^c \cup A_2^c)$,
- iv. $\Pr(A_1^c \cap A_2^c \cap A_3^c)$, and
- v. $\Pr((A_1 \cap A_2) \cup (A_2^c \cap A_3))$.

Your answers may be as brief as you wish but they must be precise. **(5pts)**

- (b) State the axioms of probability. Hence, prove formally that for any event A defined on sample space Ω , it holds that

$$\Pr(A^c) = 1 - \Pr(A),$$

where A^c denotes the complement of A . **(4pts)**

- (c) One bag contains 4 white balls and 2 black balls; another contains 3 white balls and 5 black balls. If one ball is drawn from each bag, find the probability that
- i. both are white,
 - ii. both are black,
 - iii. one is white and one is black.

You may leave your answers in terms of simplified fractions. **(6pts)**

2. Total: 15 points

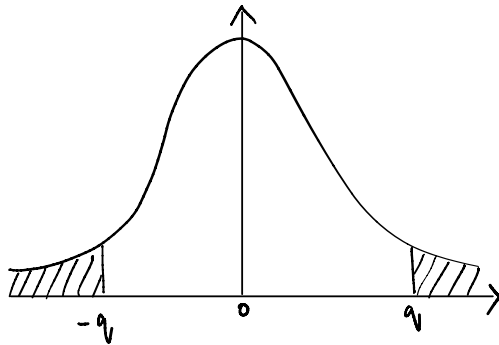
- (a) Pakistani athlete, Arshad Nadeem, set an Olympic record with a javelin throw of 92.97 metres (305.0 feet) at the 2024 Summer Olympics in Paris. Suppose we were to model the length of any one of his throws, say X , as a continuous random variable with a well-defined probability density function, say f_X . Such a model, of course, enforces that the probability that X takes any specific real value is zero. *But the javelin has to land somewhere!* Does that mean our model is useless? Explain your answer. (5pts)

- (b) Find the value of constant c such that the function

$$f_X(x) = \begin{cases} cx^2 & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

is a density function for random variable X . Hence, work out $\Pr(1 < X \leq 2)$. Evaluate the cumulative distribution function of X , say $F_X(x)$, and use $F_X(x)$ to confirm your previous answer. (5pts)

- (c) A rough sketch of Student's t distribution with 9 degrees of freedom is below.



Find the value of point $q \in \mathbb{R}^+$ (as marked on the horizontal axis) for which:

- the shaded area on the right side of the graph is 0.05,
- the total shaded area on the graph is 0.05,
- the total unshaded area on the graph is 0.99,
- the shaded area on the left of the graph is 0.01, and
- the area on the graph to the left of point q is 0.90.

(5pts)

3. Total: 10 points

- (a) Consider an Exponential(λ) random variable, X , whose density is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Also consider a Gamma(n, λ) random variable, Y , with density

$$f_Y(y) = \begin{cases} e^{-\lambda y} y^{n-1} \lambda^n / (n-1)!, & y \geq 0 \\ 0, & \text{otherwise,} \end{cases}$$

where n is a positive integer and the exclamation point denotes a factorial. Derive the moment-generating function of X , and also show that the moment-generating function of Y is given by

$$M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-n}$$

for $t < \lambda$.

(6pts)

- (b) Your colleague states that random variable $Y \sim \text{Gamma}(n, \lambda)$ may be represented as $Y = \sum_{i=1}^n X_i$, where $X_i \stackrel{iid}{\sim} \text{Exponential}(\lambda)$ for $i = 1, \dots, n$. Prove this statement using the moment-generating functions of Y and $X_1, \dots, X_n$. **(4pts)**
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4. **Total: 15 points**

- (a) Suppose $\hat{\mu}$ is an unbiased estimator of μ . You are informed that the mean-square error (MSE) of $\hat{\mu}$ is 30. What is $E(\hat{\mu}^2)$ when $\mu = 6$? **(5pts)**
- (b) Let $X_1, \dots, X_n$ denote a random sample from an Exponential($1/\theta$) distribution with density

$$f_X(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & x \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta > 0$. Derive the MSE for each of the following estimators of θ :

- i. $\hat{\theta}_1 = X_1$,
- ii. $\hat{\theta}_2 = (X_1 + X_2)/2$,
- iii. $\hat{\theta}_3 = (X_1 + 2X_2)/3$,
- iv. $\hat{\theta}_4 = (X_1 + \dots + X_n)/n$, and
- v. $\hat{\theta}_5 = 5$.

You may use (without proof) the result that for an Exponential($1/\theta$) random variable X , we have $E(X) = \theta$ and $\text{Var}(X) = \theta^2$. **(10pts)**

5. **Total: 15 points**

- (a) Suppose $X_i \stackrel{iid}{\sim} \text{Geometric}(\pi)$ for $i = 1, \dots, n$, where π is unknown. In other words, the probability mass function of each of the X_i variables is given by

$$\Pr(X_i = x) = \begin{cases} (1 - \pi)^{x-1} \pi & x = 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Derive the maximum likelihood estimator for π given a sample $\mathbf{x} = (x_1, \dots, x_n)'$. (You do **not** need to verify the second-order condition.) **(9pts)**

- (b) Consider a continuous random variable X which is uniformly distributed over the interval $[-\alpha, \alpha]$ for some $\alpha \in \mathbb{R}^+$. In other words, the density is $f_X(x) = 1/2\alpha$ for $x \in [-\alpha, \alpha]$, and 0 otherwise. Suppose we have a random sample of size n , denoted as $(x_1, \dots, x_n)$, from this distribution. Your colleague guesses that the sample mean $(1/n) \sum_{i=1}^n x_i$ yields the maximum likelihood (ML) estimate of α . Explain in words (without using equations) why your colleague's guess is likely to be incorrect. What do *you* guess will be the ML estimate of α ? **(6pts)**
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6. Total: 20 points

- (a) A machine previously produced washers having a mean thickness of 0.050 inches. To determine whether the machine is currently in proper working order, a random sample of 9 washers is chosen for which the mean thickness is 0.053 inches and the standard deviation is 0.003 inches. In other words, using conventional notation, let us denote

$$\bar{x} = (x_1 + \dots + x_9)/9 = 0.053$$

and

$$\hat{\sigma} = \sqrt{((x_1 - \bar{x})^2 + \dots + (x_9 - \bar{x})^2)/8} = 0.003.$$

Your goal is to test the hypothesis that the machine is currently in proper working order, and your colleague proposes that you use the test statistic

$$\frac{\sqrt{9}(\bar{x} - 0.050)}{\hat{\sigma}},$$

along with Student's t_8 distribution, to achieve this goal.

Provide, using your colleague's suggestion, a detailed step-by-step formulation of a test of the hypothesis that the machine is in proper working order at both 5% and 1% significance levels. Also give, with explicit reference to the statistical tables provided, your best approximation of the p -value of the test. **(15pts)**

- (b) Explain what are Type I and Type II errors in statistical hypothesis testing in any way you consider appropriate. (You may use algebra, tables/graphs, real-life examples, or any combination of these methods. Marks will be awarded for evidence of knowledge and clarity of exposition.) **(5pts)**

7. Total: 10 points

Suppose $X_t \stackrel{iid}{\sim} (\mu, \sigma_X^2)$ for $t = 1, \dots, T$ with $0 < \sigma_X^2 < \infty$ and some $T \in \mathbb{N}$.

- (a) State Chebyshev's law of large numbers, and then prove it. In your answer, you may use (without proof) Chebyshev's inequality, which states that for any random variable X with finite variance and any $\epsilon \in \mathbb{R}^+$,

$$\Pr(|X - \mathbb{E}(X)| > \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}.$$

(7pts)

- (b) State the (Lindeberg-Lévy) central limit theorem (CLT).
(You do **not** need to prove it.)

(3pts)

END OF EXAM PAPER

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z < z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576